

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Switcher
Manufacturer: Extron
Firmware Version: N/A
Model(s): SW2 HD 4K PLUS, SW4 HD 4K PLUS, SW6 HD 4K PLUS, SW8 HD 4K PLUS

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.10.0000-b007	2.6.0

Version History

Module Version	Date	Notes
1_1_5_0	9/3/2020	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly.
deviceUsername default value is admin respectively.
Example: `InterfaceName.deviceUsername = 'extron'`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='SW2 HD 4K PLUS')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='SW2 HD 4K PLUS')
```

SSHClass

#Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', 'extron'), Model='SW2 HD 4K PLUS')
```

#No Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', ''), Model='SW2 HD 4K PLUS')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value	Value	Value
AutoSwitchMode	'Off'	'User Priority'	'Input Memory Priority'
# AutoSwitchMode example InterfaceName.Set('AutoSwitchMode', 'Off')			
Command	Value	Value	
ExecutiveMode	'On'	'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command	Value	Value	
HDCPInputAuthorization	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Input'	'1' – '2' ¹ '1' – '8' ⁴	'1' – '4' ²	'1' – '6' ³
# HDCPInputAuthorization example InterfaceName.Set('HDCPInputAuthorization', 'On', {'Input': '1'})			
Command	Value	Value	Value
Input	'0' – '2' ¹ '0' – '8' ⁴	'0' – '4' ²	'0' – '6' ³
# Input example InterfaceName.Set('Input', '0')			
Command	Value	Value	Value
VideoMute	'On'	'Off'	'On with Sync'
# VideoMute example InterfaceName.Set('VideoMute', 'On')			

¹ Only available for the SW2 HD 4K PLUS model.

² Only available for the SW4 HD 4K PLUS model.

³ Only available for the SW6 HD 4K PLUS model.

⁴ Only available for the SW8 HD 4K PLUS model.

Status Available

For all commands except for AudioMute, AutoSwitchMode, ContactPort, HDCPInputAuthorization, InputSignalStatus, TallyPort, and VideoMute, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command AutoSwitchMode	Value 'Off'	Value 'User Priority'	Value 'Input Memory Priority'
# AutoSwitchMode example InterfaceName.Update('AutoSwitchMode') Value = InterfaceName.ReadStatus('AutoSwitchMode') InterfaceName.SubscribeStatus('AutoSwitchMode', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ContactPort	Value 'Open'	Value 'Closed'	
Qualifier Key 'Port'	Qualifier Value '1' – '2' ¹ '1' – '8' ⁴	Qualifier Value '1' – '4' ²	Qualifier Value '1' – '6' ³
# ContactPort example InterfaceName.Update('ContactPort', {'Port': '1'}) Value = InterfaceName.ReadStatus('ContactPort', {'Port': '1'}) InterfaceName.SubscribeStatus('ContactPort', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command HDCPInputAuthorization	Value 'On'	Value 'Off'	
Qualifier Key 'Input'	Qualifier Value '1' – '2' ¹	Qualifier Value '1' – '4' ²	Qualifier Value '1' – '6' ³

	'1' – '8' ⁴		
# HDCPInputAuthorization example InterfaceName.Update('HDCPInputAuthorization', {'Input': '1'}) Value = InterfaceName.ReadStatus('HDCPInputAuthorization', {'Input': '1'}) InterfaceName.SubscribeStatus('HDCPInputAuthorization', None, FeedbackHandler)			
Command Input	Value '0' – '2' ¹ '0' – '8' ⁴	Value '0' – '4' ²	Value '0' – '6' ³
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command InputSignalStatus	Value 'Active'	Value 'Not Active'	
Qualifier Key 'Input'	Qualifier Value '1' – '2' ¹ '1' – '8' ⁴	Qualifier Value '1' – '4' ²	Qualifier Value '1' – '6' ³
# InputSignalStatus example InterfaceName.Update('InputSignalStatus', {'Input': '1'}) Value = InterfaceName.ReadStatus('InputSignalStatus', {'Input': '1'}) InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)			
Command TallyPort	Value 'Open'	Value 'Closed'	
Qualifier Key 'Port'	Qualifier Value '1' – '2' ¹ Qualifier Value '1' – '8' ⁴	Qualifier Value '1' – '4' ²	Qualifier Value '1' – '6' ³
# TallyPort example InterfaceName.Update('TallyPort', {'Port': '1'}) Value = InterfaceName.ReadStatus('TallyPort', {'Port': '1'}) InterfaceName.SubscribeStatus('TallyPort', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	Value 'On with Sync'
# VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			

¹ Only available for the SW2 HD 4K PLUS model.² Only available for the SW4 HD 4K PLUS model.³ Only available for the SW6 HD 4K PLUS model.⁴ Only available for the SW8 HD 4K PLUS model.

Cable and Adapter Requirements

Captive Screw to Captive Screw RS-232 Serial Cable.

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

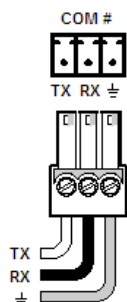
Data Bits: 8

Parity: None

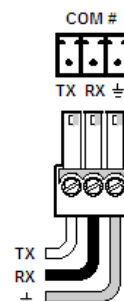
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Signal
TxD		TxD
RxD		RxD
GND		GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 22023

Logon Credentials Supported: Yes

Default Username: admin

Default Password:

Multi-Connection Yes

Capabilities:

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Audio Mute Off	w0AFMT\x0D
Audio Mute On	w1AFMT\x0D
Auto Switch Mode Input Memory Priority	w2AUSW\x0D
Auto Switch Mode Off	w0AUSW\x0D
Auto Switch Mode User Priority	w1AUSW\x0D
Executive Mode Off	0X
Executive Mode On	1X
HDCP Input Authorization Off Input 1	wE1*0HDCP\x0D
HDCP Input Authorization Off Input 2	wE2*0HDCP\x0D
HDCP Input Authorization Off Input 4	wE4*0HDCP\x0D
HDCP Input Authorization Off Input 6	wE6*0HDCP\x0D
HDCP Input Authorization Off Input 8	wE8*0HDCP\x0D
HDCP Input Authorization On Input 1	wE1*1HDCP\x0D
HDCP Input Authorization On Input 2	wE2*1HDCP\x0D
HDCP Input Authorization On Input 4	wE4*1HDCP\x0D
HDCP Input Authorization On Input 6	wE6*1HDCP\x0D
HDCP Input Authorization On Input 8	wE8*1HDCP\x0D
Input 0	0!
Input 2	2!
Input 4	4!
Input 6	6!
Input 8	8!
Video Mute Off	0B
Video Mute On	1B
Video Mute On with Sync	2B

Appendix B. Update Commands

Audio Mute, Auto Switch Mode, Input, Video Mute	I
Contact Port Port	S
Executive Mode	X
HDCP Input Authorization	wEHDCP\x0D
Input Signal Status	wLS\x0D
Tally Port	wTALY\x0D

